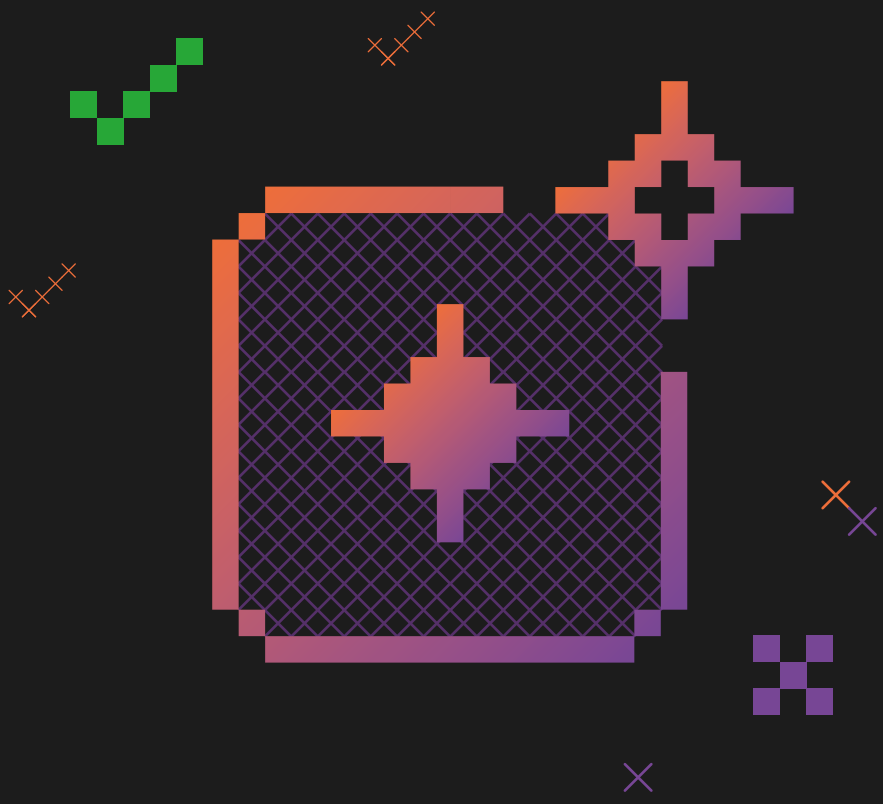
XRAY

AI in Testing: What every QA Team needs to know



Jira

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Introduction: AI in Software Testing

Artificial Intelligence is no longer a distant trend — it is already influencing the way software is built, deployed, and tested. In quality assurance, the conversation around AI has grown louder as teams search for ways to keep pace with rapid release cycles and rising expectations for reliability.

Still, **there's a gap between speculation and reality**. Headlines often promise fully autonomous testing or instant defect elimination, **but the truth is more nuanced**. Some applications of AI in testing are already practical and valuable, while others remain aspirational. For QA teams, separating hype from tangible progress is the first step toward making **informed decisions**.

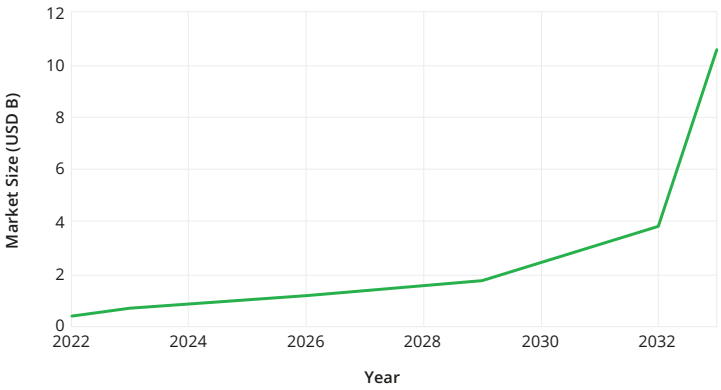
What matters most is perspective. **AI is not here to replace testers or diminish their role**. Instead, it can extend their capabilities by handling repetitive, data-heavy tasks, uncovering patterns at scale, and providing insights that support smarter strategies. Human testers bring context, judgment, and creativity — qualities AI cannot replicate. When these strengths are combined, teams can reach higher levels of efficiency and quality.

Xray's mission is to enable organizations to achieve excellence in testing through structure, speed, and collaboration. As AI reshapes the software landscape, Xray's role is to help teams navigate this evolution with clarity and confidence. This eBook explores where AI can make a real impact, the challenges that come with adoption, and how QA teams can prepare for the future without losing sight of their core purpose: delivering quality software that users trust. **Xray's AI capabilities are here to support you.**

Understanding AI in Testing

01

Forecast Growth of AI in Software Testing Market (2022-2033)



AI has become a catch-all phrase in the industry, but in the context of software testing, it's essential to break it down into what it really means and what it can actually do today.
At its core, Artificial Intelligence refers to systems that can perform tasks requiring human intelligence (pattern recognition, prediction, or decision-making). Within this umbrella, several areas are relevant for QA:

1. **Machine Learning (ML):** Algorithms that learn from data and improve over time. In testing, this can mean predicting defect-prone areas of code or identifying patterns in execution failures.

2. **Generative AI (GenAI):** Models that create new content, from natural language text to code snippets. For QA teams, this translates into support for writing test cases, generating test data, or refining acceptance criteria.

3. **Cognitive Automation:** Systems designed to mimic reasoning and problem-solving. In testing, this might be applied to classify defects, cluster logs, or recommend test coverage improvements.

Understanding these distinctions is key. Not all AI is equal, and not every claim of "AI-powered testing" represents the same level of maturity. Today, practical use cases exist — AI copilots that assist with documentation, test data generation, or bug triage. At the same time, more ambitious ideas such as fully autonomous test maintenance or end-to-end self-healing pipelines remain limited to specific tools and contexts.

By clarifying capabilities and limitations, QA leaders can set realistic expectations. AI can already drive efficiency in supporting activities, but it cannot replace human oversight or strategic decision-making. Credibility comes from knowing where AI can add immediate value and where its role is still evolving.

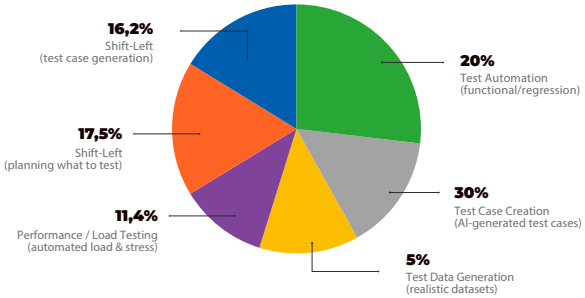
For Xray, adding value to your team matters and our focus is helping teams prepare for change without overpromising outcomes. AI should be seen as an external enabler that complements established testing practices rather than as a disruptive force that invalidates them.

Impact of AI Across Testing Practices

02

AI's impact in testing is best understood not as a single innovation but as a collection of targeted applications across disciplines. Each type of testing faces unique challenges, and AI can provide tailored support that complements human expertise while reducing repetitive effort.

% of QA Teams Using AI per Area



Regarding exploratory testing not having quantified results:

Exploratory testing is not included in the chart because no large-scale survey data currently exists on AI adoption in this area. Early reports, however, point to teams experimenting with AI for tasks such as summarizing notes, clustering defects, and highlighting risk areas.

Exploratory Testing

Exploratory testing is dynamic by nature. Testers follow their intuition, experiment with scenarios, and capture insights in real time. While this approach uncovers issues traditional scripted tests may miss, it also produces fragmented artifacts (notes, screenshots, bug reports) that can be difficult to manage or analyze systematically.

AI can transform this process by acting as a silent partner. What does this mean? Natural language processing (NLP) models can summarize tester notes into structured reports, ensuring that valuable observations are not lost. Similarly, clustering algorithms can group related defects by similarity, pointing testers toward systemic issues rather than isolated cases.

Another promising area is **risk-based exploration guidance**. By analyzing past defects, recent code changes, or production logs, AI can suggest areas that deserve closer investigation. Rather than replacing tester creativity, this acts as a compass, helping teams prioritize where to spend their limited time.

The result is exploratory sessions that are both richer in insights and easier to share, enhancing collaboration between testers, developers, and stakeholders.

Shift-Left Testing

Shifting quality activities earlier in the development lifecycle reduces cost and rework. Yet, early testing requires reliable indicators of where defects are likely to emerge, which AI is well suited to provide.

For instance, machine learning models trained on historical project data can predict defect-prone modules based on patterns in code complexity, commit frequency, or developer activity. AI can enhance static code analysis by learning which warnings historically led to real bugs, filtering out noise, and prioritizing the most relevant alerts.

AI can also analyze requirements written in natural language, flagging ambiguities or contradictions before they enter development. In a shift-left strategy, **this strengthens collaboration** between business analysts, developers, and testers, ensuring quality is built in from the start.

The practical outcome is earlier detection of defects, but more importantly, better alignment of requirements, code, and tests.

Test Case Design and optimization

Designing effective test suites is one of the most resource-intensive parts of QA. As systems evolve, suites often grow bloated, filled with redundant or outdated cases that slow execution and add maintenance overhead. AI can bring much-needed efficiency to this process.

Generative AI can assist testers by drafting initial test cases from user stories or acceptance criteria. While human review remains essential, this reduces the time spent on routine authoring. Beyond creation, clustering and similarity detection models can identify overlapping tests.

AI is also valuable in **test data generation**. Instead of relying on static datasets, testers can use AI to create varied and realistic inputs, broadening coverage and uncovering edge cases that might otherwise be missed.

The result is not simply "more tests," but leaner, smarter suites that **deliver higher confidence with less overhead**. Over time, this shifts the tester's role from writing large volumes of cases to curating and refining a strategic test portfolio.

Test Automation & maintenance

Automation promises speed, but its biggest challenge is sustainability. Scripts frequently break when applications change (it can go from new UI elements to updated APIs or altered workflows). Maintenance can consume more effort than the initial creation, limiting the scalability of automation.

AI addresses this by introducing **resilience**. Self-healing scripts use AI to **detect when an element has changed and automatically adjust to the new identifier**. This reduces the constant cycle of broken tests and manual fixes. Visual validation powered by image recognition can detect UI regressions that traditional assertions might miss, providing a more user-centric measure of correctness.

AI can also support test execution optimization. Instead of running entire suites, AI models can analyze recent code changes and historical failures to recommend the most relevant subset of tests. This accelerates CI/CD pipelines without sacrificing coverage.

While these approaches are still evolving, they point toward a future where automation becomes faster, smarter, and more adaptive, freeing teams from repetitiveness and allowing them to focus on higher-value tasks.

Performance & Load Testing

Performance testing has always been about anticipating how systems behave under stress. The challenge lies in simulating realistic workloads and interpreting complex results. AI brings new capabilities that move teams closer to predictive performance engineering.

When using AI in Performance Testing, you can model authentic user journeys by analyzing production telemetry. You can replace generic load profiles with simulations that mirror actual behavior, ensuring that performance tests reflect real-world usage and increasing their relevance and accuracy.

AI can also detect anomalies in performance trends before they escalate. Predictive models can learn from historical data to forecast where degradation is likely to occur. Instead of reacting to performance failures, teams can prevent them.

This shift from detection to prediction transforms performance testing from a late-stage activity to a **continuous, intelligence-driven practice**. Ultimately, it helps organizations deliver systems that are resilient to growth and adaptable to user demand.

Across all of these domains, the theme is consistent: **AI amplifies the strengths of QA teams rather than displacing them**. Each application provides opportunities to reduce manual effort, improve accuracy, and align testing more closely with business risk. What remains essential is human judgment: directing where AI is applied, interpreting its outputs, and ensuring that **quality remains a strategic goal rather than a byproduct of automation**.

Ethical Considerations and Organizational Readiness

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The adoption of AI in testing is not just a technical shift — it also raises ethical questions and organizational challenges that cannot be ignored. If you move too fast without addressing these areas, you risk losing trust and hurting long-term results.

Data privacy and security

AI systems often require large volumes of data to function effectively. In testing, this may include logs, user interactions, or production telemetry. Without safeguards, **sensitive information could be exposed**. QA leaders must ensure that data used for training or prompting AI models complies with privacy regulations such as GDPR or HIPAA, and that anonymisation practices are in place. **Protecting user trust is not optional is a baseline requirement**.

This is where compliance-driven testing becomes essential. Solutions like Xray Enterprise support QA teams by providing end-to-end traceability, electronic approvals, and audit-ready reporting, making it easier to align AI testing processes with strict regulatory frameworks. By embedding compliance into testing workflows, organizations can innovate with AI while safeguarding both user data and regulatory integrity (learn how Xray Enterprise simplifies compliance).

Bias and accountability

AI models learn from historical data, and if that data contains bias, the outputs will reflect it. In testing, this could manifest as skewed defect prioritisation or overlooked edge cases. While AI can highlight risks, final accountability for decisions must remain with humans. A human-in-the-loop approach, where testers validate AI-driven recommendations before acting, ensures that judgment and context are never removed from the process.

Impact on tester roles

Another concern is the perception that AI may replace testers. In reality, AI is far better suited to repetitive or large-scale data analysis than to the nuanced, creative work of designing test strategies or interpreting ambiguous requirements. Still, roles will evolve. Testers will need to become proficient in interpreting AI outputs, spotting when recommendations are useful and when they miss the mark. This requires new skills, but it also opens opportunities for professional growth.

Upskilling and change management

Organizational readiness depends not just on tools but on people. Introducing AI into testing workflows requires:

1. **Skills development** – Testers may need training in prompt engineering, critical evaluation of AI outputs, and awareness of ethical risks.

2. **Change management** – Leadership needs to communicate clearly why AI is being introduced, how it will support rather than replace teams, and what guardrails are in place.

3. **Risk frameworks** – Organizations should develop policies for where AI can and cannot be applied, ensuring consistency and trust in its use.

Responsible adoption

Responsible adoption balances ambition with caution. It means piloting AI on narrow, low-risk use cases before scaling, maintaining transparency around AI usage, and continuously reviewing outcomes. Organizations can harness AI's benefits by embedding responsibility into the process without compromising ethics, compliance, or team confidence.

Ultimately, success with AI in testing will not come from technology alone. It will come from organizations that are prepared to use AI as a supportive partner in their pursuit of quality.

AI as an assistant for QA Teams

04

AI as an assistant for QA Teams

The most practical way to think about AI in testing today is as a partner that supports, rather than replaces. Instead of aiming for full autonomy, AI adds value by removing repetitive burdens, streamlining workflows, and enhancing productivity within established processes.

AI copilots for daily QA activities

One of the clearest opportunities lies in test authoring and documentation. Drafting test cases, acceptance criteria, or compliance checklists is time-consuming and often repetitive. Generative AI models can provide a starting point by translating requirements into draft scenarios, or by turning exploratory session notes into structured documentation. This does not remove the need for human oversight, but it reduces the manual load so testers can focus on refining.

In regulated industries, where compliance validation is crucial, **AI can assist by scanning test documentation for missing information, inconsistencies, or formatting errors**. Acting as a quality gate, AI ensures documentation is complete and consistent before it moves forward.

Integration into toolchains

AI's usefulness increases when embedded directly into the tools teams already rely on. For example:

- Within Jira, AI copilots could assist in refining user stories or linking them to appropriate test cases.
- In Confluence, they could generate concise summaries of long requirement documents or test strategy outlines.
- Within CI/CD pipelines, AI could highlight unstable tests, recommend execution subsets, or flag build failures with likely causes.

These integrations do not reinvent workflows. They augment existing practices, providing testers with actionable insights without forcing tool switching or additional overhead.

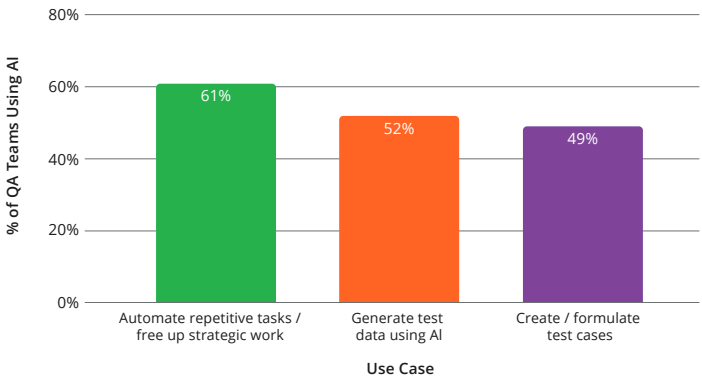
Practical example: LLM-driven acceptance criteria refinement

Consider a behavior-driven development (BDD) context. Writing Given/When/Then scenarios requires precision, but requirements are often expressed in ambiguous natural language. A large language model (LLM) can analyze draft acceptance criteria, suggest refinements, and highlight edge cases the author may not have considered. The human tester or product owner retains control, but the process is faster, clearer, and less error-prone.

Shaping a collaborative mindset

By positioning AI as a copilot rather than a replacement, QA leaders can reduce resistance to adoption. Testers remain in the driver's seat and AI accelerates the journey. This framing is crucial for building trust within teams, ensuring that AI adoption feels like an opportunity for professional growth, not a threat to job security.

AI in QA: Top Areas of Adoption



Emerging areas and future trends

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AI in software testing is still developing, but several frontiers show how the discipline may evolve in the coming years. Awareness of these trends is more about preparing their teams to recognize opportunities and challenges as they mature.

AI-powered risk-based testing at scale

Risk-based testing is not new; it prioritizes test effort on features with the highest potential business impact. What AI adds is the ability to analyze vast datasets and generate dynamic risk profiles in real time. Instead of relying on manual risk assessments conducted once per release, AI-driven models could continuously recommend which product areas deserve the most testing attention.
This brings efficiency: teams spend less time running exhaustive test suites and more time focusing on high-value testing aligned to business risk. Over time, this could redefine regression strategies, shifting from "run everything" to "run what matters most right now."

Integration with DevOps observability

As software delivery accelerates, the lines between development, operations, and testing blur. Observability platforms already collect vast streams of logs, metrics, and traces from production systems. AI has the potential to connect this operational data back into the QA process.

For example, anomalies in production usage could automatically trigger test recommendations. AI could generate targeted performance or integration tests if an observability platform detects that a particular API is slowing under specific conditions. This closes the loop between real-world behavior and pre-release validation, **making testing more proactive and context-driven**.

AI in MLOps (Machine Learning Operations) and testing AI systems

As organizations adopt AI and machine learning in their own products, QA teams face a new responsibility: testing AI models themselves. This introduces challenges such as bias detection, fairness validation, and monitoring model drift. Emerging practices in MLOps show how AI systems require ongoing validation, not just one-time testing.

Here, AI can support testers by analyzing model outputs for inconsistencies, identifying patterns of bias, or simulating edge cases to test robustness. This domain is still in its early stages, but it shows how testing as a profession will expand in scope alongside the systems it validates.

Strategic foresight

These emerging areas may not be part of every QA team's workflow today, but they signal where the discipline is heading. By staying informed and preparing for gradual adoption, teams can position themselves ahead of the curve.

For Xray, the priority is not to promise solutions that are not yet mature, but to act as a **trusted partner for QA leaders navigating change**. By providing clarity on realistic applications and future trends, we aim to equip teams with the knowledge they need to make confident, long-term decisions about AI in testing.

Navigating the AI Testing Evolution with Confidence

06

Navigating the AI Testing evolution with confidence

AI is reshaping the landscape of software testing. From Exploratory Testing to automation, from performance engineering to compliance, the opportunities are real, and so are the challenges. The key is perspective. AI should be seen not as a disruptive force that undermines the role of testers, but as a strategic assistant that augments human expertise and brings clarity to complex data.

The journey forward requires balance. Teams must learn to distinguish hype from tangible value, adopt AI responsibly, and build readiness through skills, mindset, and leadership. Ethical considerations, from data privacy to bias, make AI adoption sustainable and trustworthy. At the same time, forward-looking teams can already begin to capture value by piloting AI in targeted areas, creating space for creativity, strategy, and collaboration.

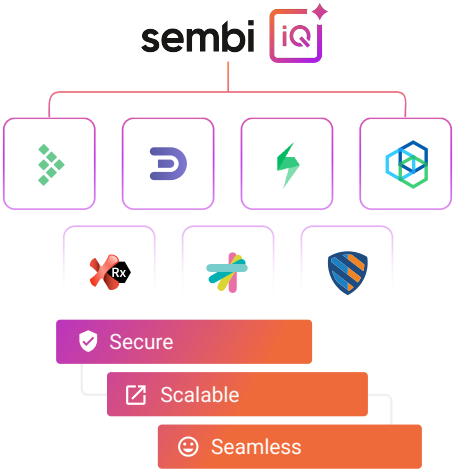
Xray's role is to help QA teams face this evolution with confidence. **Our mission remains unchanged: to empower organizations to deliver excellence in quality**, supported by structure, security, and insight. What changes with AI is the toolkit available to achieve that mission.

Introducing Xray's new AI-Powered capabilities

07

To support teams on this journey, we have launched two new AI Capabilities designed to accelerate testing while keeping control firmly in the hands of testers. Both are powered by Sembi IQ, Sembi's AI platform that infuses intelligence across the entire software lifecycle.

What is Sembi IQ?



Sembi IQ is the intelligence layer that powers Xray's AI capabilities. AI features are opt-in only, ensuring customers stay in full control. Data security is built in from the ground up: customer data is always isolated, never shared across instances, and never used for model training.

Rooted in software testing and security, Sembi IQ enhances how teams build and test software. It accelerates test creation with security-minded scenarios while helping QA, development, and security teams deliver safer, higher-quality products.

By making testing more **contextual, explainable, and connected**, Sembi IQ helps organizations release better software—faster and more securely.

For Xray users, that means:

- AI that works natively within Jira, enhancing the workflows you already rely on.
- Intelligence that improves decision-making without introducing extra tools or complexity.
- Features are designed to guide — not replace — testers, keeping oversight and accountability where they belong.

For Sembi IQ, **your trust and security are priority**. AI features are not turned on by default—customers must opt in. Customer data is kept entirely separate, with no leakage or mixing between instances. Data is never used for model training, ensuring maximum security.

Sembi is rooted in software testing and security. AI features not only help you test faster, but also empower you to build safer, more secure products. Test cases are designed with security-minded scenarios.

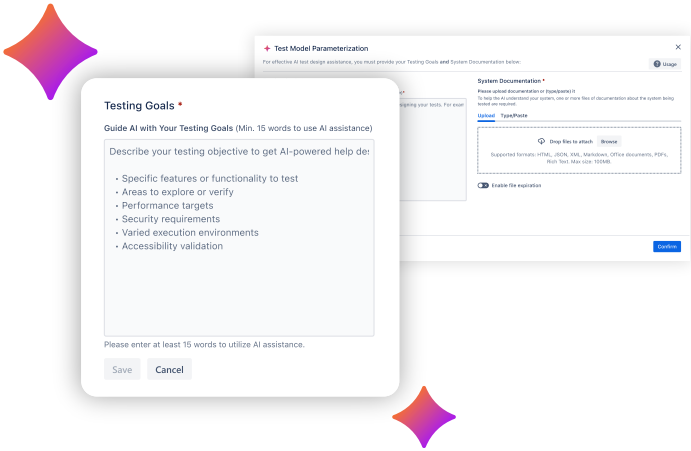
With Sembi IQ, Xray is moving beyond automation into a future where testing is smarter, faster, and aligned with the real challenges QA teams face.

AI-Powered Test Case Generation (Xray Standard, Advanced, and Enterprise)

Turn your requirements into actionable test cases in seconds — without sacrificing oversight.

- Instantly generate test cases directly from requirements.
- Review, edit, and maintain complete control of outputs.
- Create both Manual and BDD (Cucumber) test formats.
- Built with security, privacy, and scalability at its core.

AI Test Model Generation (Xray Enterprise)



Powered by AI, this feature accelerates one of the most critical steps in modeling: defining Parameters and Values.

- Quickly generate initial Parameters and Values for your models.
- Expand or refine values to fine-tune coverage.
- Leverage advanced techniques like boundary testing and equivalence classes.
- Provide high-level goals, and let AI adapt as your model evolves.

As AI continues to evolve, so will Xray. Our vision is to further empower QA teams with intelligent, context-aware capabilities that enhance productivity and simplify complex testing workflows.

We're looking ahead to a future where AI not only accelerates test creation and modeling but also supports automation and decision-making across every stage of the testing process. These upcoming capabilities represent the next step in that journey — expanding how Xray helps teams deliver quality software, faster and with greater confidence.

AI Automated Script Generation

Transform manual tests into automated scripts — seamlessly and intelligently. Powered by **Sembi IQ**, this feature bridges the gap between manual and automated testing, helping teams accelerate automation without leaving Xray.

- Converts existing **manual test cases** into executable automation scripts in supported frameworks.
- Ensures scripts align with project context and maintain full traceability to requirements.
- Reduces manual scripting effort while empowering testers with limited coding experience to contribute to automation.
- Integrates with existing CI/CD pipelines and automation frameworks for a consistent, end-to-end workflow.
- Helps teams scale automation faster, with less effort and higher accuracy.

AI Test Prioritization

Optimize your test executions by focusing on what matters most. Powered by **Sembi IQ**, this capability analyzes project data to intelligently recommend which tests to run first — maximizing efficiency while maintaining coverage.

- Leverages context from requirements, test cases, historical defects, and past executions to identify high-value tests.
- Automatically assigns prioritization levels (e.g., critical, high, medium) with short explanations for each suggestion.
- Allows users to review, adjust, and create Test Plans or Test Executions based on AI recommendations.
- Supports flexible prioritization, letting users focus on recent defects, execution history, or specific release types.
- Helps teams accelerate releases by testing less while covering more.

In all of Xray's AI features, AI acts as a guide, not a replacement. Testers remain in control — AI offers suggestions, and you decide what to accept.

The evolution of AI in testing is only beginning, but with the right approach, it offers teams a powerful way to scale quality without losing control. With Sembi IQ as the backbone, Xray is delivering a new level of intelligence that helps teams move faster without compromising quality. Xray will continue to lead with clarity, responsibility, and innovation, ensuring that QA professionals are equipped not only for today's challenges but also for tomorrow's opportunities.

Links

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